Calcurion Maths Worksheets

A-Level Mathematics

Topic: Pure 2 – Differential Equations

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Solve $\frac{dy}{dx} = \frac{2y}{x+1}$, $y(0) = 4$. **[5]**

2. Solve $\frac{dy}{dx} = 3y$, $y(0) = 2$. **[4]**

3. Solve $\frac{dy}{dx} = 2xy$, $y(0) = 1$. **[5]**

4. Solve $\frac{dy}{dx} = \frac{y}{x}$, $y(1) = 3$. **[4]**

5. Solve $\frac{dy}{dx} = x(y + 1)$. General solution. **[4]**

6. Solve $\frac{dy}{dx} = \frac{y+1}{x}$. General solution. **[4]**

7. Solve $\frac{dy}{dx} = e^{x-y}$. General solution. **[5]**

8. Solve $\frac{dy}{dx} = \frac{x}{y}$, $y(0) = 2$. **[5]**

9. Solve $\frac{dy}{dx} = y^2$, $y(0) = 1$. **[5]**

10. Solve $\frac{dy}{dx} + 2y = 0$, $y(0) = 5$. **[4]**

11. Solve $\frac{dy}{dx} - y = 0$, $y(1) = e$. **[4]**

12. Solve $\frac{dy}{dx} = \frac{2x}{y}$, $y(1) = 2$. **[5]**

13. Solve $\frac{dy}{dx} = 3x^2 y$, $y(0) = 2$. **[5]**

14. Solve $\frac{dy}{dx} = \frac{y}{x^2}$, $y(1) = e$. **[5]**

15. Solve $\frac{dy}{dx} + y = 0$. General solution. **[3]**

Section B – Hard (Q16–25)

16. Solve $\frac{dy}{dx} + 2y = e^{-x}$, $y(0) = 1$. **[7]**

17. Solve $\frac{dy}{dx} + y = x$, $y(0) = 0$. **[7]**

18. Solve $\frac{dy}{dx} - 3y = e^{2x}$, $y(0) = 1$. **[7]**

19. Population P: $\frac{dP}{dt} = 0.05P$, $P(0) = 500$. Find $P(t)$ and time when $P = 2000$. **[6]**

20. Population P: $\frac{dP}{dt} = 0.03P$, $P(0) = 1000$. Find P after 20 years. **[5]**

21. Solve $\frac{dy}{dx} = \frac{y}{x+1} + (x+1)$, $y(0) = 1$. **[7]**

22. Solve $\frac{dy}{dx} + xy = x$, $y(0) = 2$. **[6]**

23. The rate of cooling satisfies $\frac{d\theta}{dt} = -k\theta$. Given $\theta(0) = 80$, $\theta(10) = 40$, find $\theta(t)$. **[6]**

24. Solve $\frac{dy}{dx} = \frac{2y}{x} + x$, $y(1) = 2$. **[6]**

25. Solve $\frac{dy}{dx} = y\cot x$, $y(\pi/2) = 1$. **[5]**

End of Worksheet • *Show all working clearly.*